

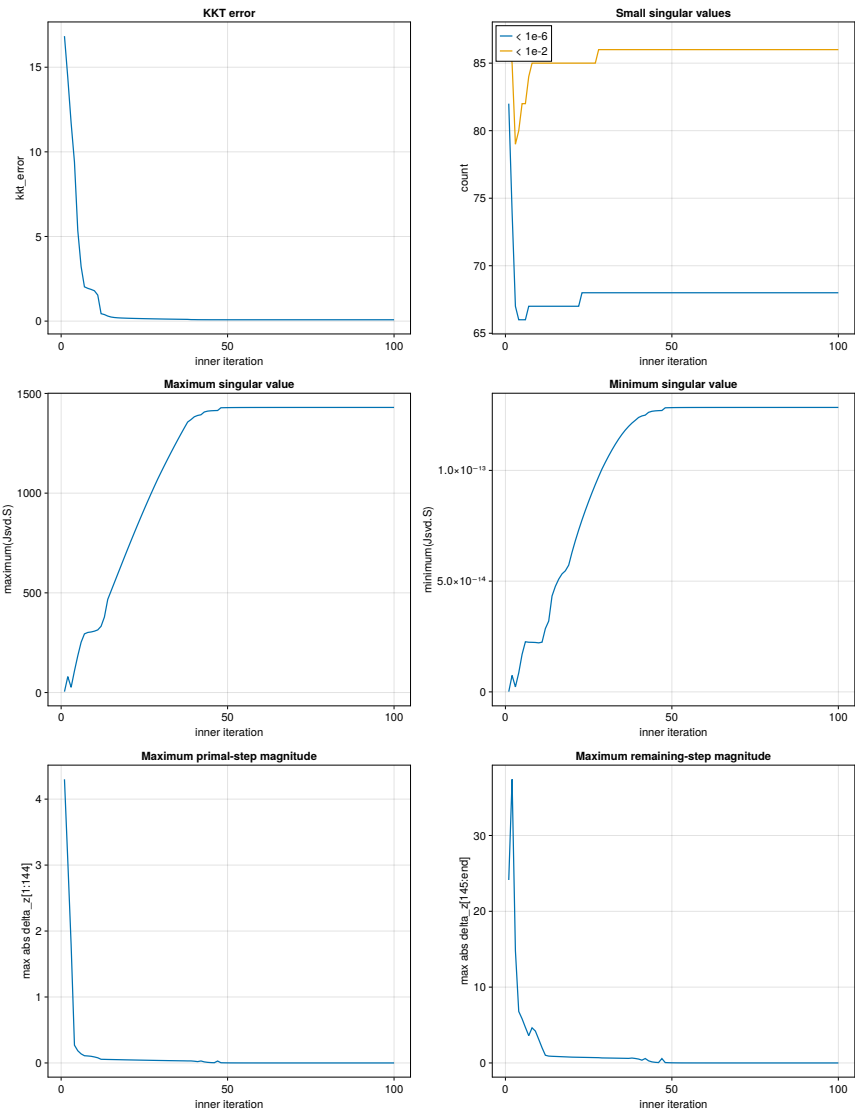
Solver Diagnostics

Recorded 100 inner iterations; reporting 100 rows (cap: 1000).

Summary statistics

Metric	Minimum	Maximum	Mean	Median
kkt_error	7.919147e-02	1.683624e+01	7.962595e-01	7.919351e-02
count(Jsvd.S < 1e-6)	6.600000e+01	8.200000e+01	6.797000e+01	6.800000e+01
count(Jsvd.S < 1e-2)	7.900000e+01	8.700000e+01	8.557000e+01	8.600000e+01
maximum(Jsvd.S)	4.591747e+00	1.429746e+03	1.147772e+03	1.428902e+03
minimum(Jsvd.S)	1.118702e-16	1.284201e-13	1.032452e-13	1.283431e-13
maximum(abs.(delta_z[1:144]))	1.055097e-07	4.300357e+00	1.155498e-01	1.096398e-03
maximum(abs.(delta_z[145:end]))	1.610403e-06	3.740878e+01	1.343731e+00	1.613357e-02

Metrics versus inner iteration



Per-iteration diagnostics

Iter	KKT error	$N_{<10^{-6}}$	$N_{<10^{-2}}$	$\sigma_{\max}$	$\sigma_{\min}$	$\max \delta z_{1:144} $	$\max \delta z_{145:} $
1	1.683624e+01	82	87	4.591747e+00	1.118702e-16	4.300357e+00	2.413179e+01
2	1.437564e+01	74	85	7.997262e+01	7.436759e-15	3.064875e+00	3.740878e+01
3	1.171038e+01	67	79	2.638581e+01	2.244389e-15	1.792522e+00	1.479727e+01
4	9.397956e+00	66	80	1.086637e+02	8.617732e-15	2.693512e-01	6.793148e+00
5	5.352789e+00	66	82	1.843782e+02	1.691558e-14	1.847745e-01	5.826263e+00
6	3.223161e+00	66	82	2.522099e+02	2.263070e-14	1.374185e-01	4.690754e+00
7	2.025214e+00	67	84	2.941899e+02	2.241171e-14	1.095991e-01	3.600391e+00
8	1.936404e+00	67	85	3.013522e+02	2.237477e-14	1.058524e-01	4.633127e+00
9	1.872177e+00	67	85	3.037981e+02	2.230935e-14	1.009111e-01	4.206068e+00
10	1.792011e+00	67	85	3.078606e+02	2.213370e-14	9.094990e-02	3.117907e+00
11	1.537073e+00	67	85	3.141684e+02	2.243926e-14	7.822695e-02	2.006534e+00
12	4.388336e-01	67	85	3.324143e+02	2.858418e-14	5.538985e-02	1.014786e+00
13	3.860894e-01	67	85	3.788744e+02	3.196471e-14	5.433192e-02	9.033996e-01
14	2.978419e-01	67	85	4.670545e+02	4.329667e-14	5.294822e-02	8.767465e-01
15	2.410071e-01	67	85	5.093555e+02	4.763637e-14	5.194952e-02	8.592631e-01
16	2.116739e-01	67	85	5.517626e+02	5.088076e-14	5.090859e-02	8.411935e-01
17	1.948820e-01	67	85	5.941339e+02	5.327081e-14	4.984796e-02	8.228801e-01
18	1.839621e-01	67	85	6.363463e+02	5.465852e-14	4.876702e-02	8.043430e-01
19	1.762591e-01	67	85	6.786123e+02	5.720227e-14	4.757115e-02	7.835487e-01
20	1.691255e-01	67	85	7.202725e+02	6.299645e-14	4.655248e-02	7.661076e-01
21	1.632570e-01	67	85	7.614986e+02	6.823906e-14	4.554253e-02	7.568905e-01
22	1.580664e-01	67	85	8.022292e+02	7.308096e-14	4.454533e-02	7.479361e-01
23	1.532860e-01	68	85	8.424124e+02	7.761763e-14	4.356346e-02	7.389646e-01
24	1.487879e-01	68	85	8.820047e+02	8.190937e-14	4.258015e-02	7.310719e-01
25	1.444317e-01	68	85	9.209807e+02	8.599097e-14	4.163652e-02	7.218990e-01
26	1.403193e-01	68	85	9.593029e+02	8.989070e-14	4.071066e-02	7.127047e-01
27	1.363918e-01	68	85	9.969472e+02	9.362240e-14	3.980310e-02	7.034922e-01
28	1.326219e-01	68	86	1.033895e+03	9.719361e-14	3.891407e-02	6.942638e-01
29	1.289947e-01	68	86	1.070133e+03	1.006073e-13	3.795226e-02	6.614237e-01
30	1.250127e-01	68	86	1.104300e+03	1.036269e-13	3.712774e-02	6.551765e-01
31	1.216964e-01	68	86	1.137940e+03	1.065037e-13	3.631451e-02	6.482380e-01
32	1.185922e-01	68	86	1.171010e+03	1.092218e-13	3.551455e-02	6.408953e-01
33	1.155864e-01	68	86	1.203483e+03	1.117658e-13	3.472880e-02	6.332842e-01
34	1.126583e-01	68	86	1.235341e+03	1.141197e-13	3.395781e-02	6.254785e-01
35	1.098088e-01	68	86	1.266574e+03	1.162660e-13	3.320187e-02	6.175243e-01
36	1.070414e-01	68	86	1.297173e+03	1.181856e-13	3.246115e-02	6.094546e-01
37	1.043578e-01	68	86	1.327135e+03	1.198580e-13	3.173574e-02	6.012948e-01
38	1.017580e-01	68	86	1.356458e+03	1.213497e-13	3.063555e-02	6.459621e-01
39	8.924905e-02	68	86	1.368672e+03	1.225973e-13	3.067083e-02	5.904005e-01
40	8.768453e-02	68	86	1.382863e+03	1.238933e-13	2.635583e-02	5.126079e-01
41	8.412504e-02	68	86	1.389492e+03	1.245815e-13	2.029923e-02	3.667354e-01
42	8.253747e-02	68	86	1.392948e+03	1.249522e-13	2.997455e-02	5.850704e-01
43	8.193933e-02	68	86	1.406902e+03	1.262082e-13	1.607299e-02	2.803172e-01
44	8.140186e-02	68	86	1.411222e+03	1.266773e-13	9.129758e-03	1.420170e-01
45	8.013134e-02	68	86	1.412943e+03	1.268719e-13	5.362007e-03	8.258018e-02
46	7.998651e-02	68	86	1.413833e+03	1.269736e-13	3.302149e-03	5.072034e-02
47	7.996502e-02	68	86	1.414374e+03	1.270342e-13	2.924208e-02	5.798491e-01
48	7.956221e-02	68	86	1.428122e+03	1.282526e-13	3.117524e-03	4.913937e-02
49	7.919784e-02	68	86	1.428480e+03	1.282968e-13	1.997034e-03	3.056465e-02
50	7.919420e-02	68	86	1.428794e+03	1.283317e-13	1.277482e-03	1.954677e-02
51	7.919283e-02	68	86	1.429009e+03	1.283546e-13	9.153135e-04	1.272037e-02
52	7.919224e-02	68	86	1.429166e+03	1.283705e-13	6.748978e-04	8.389086e-03
53	7.919194e-02	68	86	1.429283e+03	1.283817e-13	5.080389e-04	5.611634e-03
54	7.919177e-02	68	86	1.429372e+03	1.283899e-13	3.897119e-04	4.224001e-03
55	7.919167e-02	68	86	1.429441e+03	1.283960e-13	3.040444e-04	3.543284e-03
56	7.919161e-02	68	86	1.429495e+03	1.284007e-13	2.407384e-04	2.975674e-03
57	7.919157e-02	68	86	1.429539e+03	1.284043e-13	1.930285e-04	2.501141e-03
58	7.919154e-02	68	86	1.429574e+03	1.284071e-13	1.564063e-04	2.103664e-03
59	7.919152e-02	68	86	1.429603e+03	1.284094e-13	1.278241e-04	1.770252e-03
60	7.919151e-02	68	86	1.429626e+03	1.284112e-13	1.051886e-04	1.490275e-03
61	7.919150e-02	68	86	1.429646e+03	1.284127e-13	8.703713e-05	1.254970e-03
62	7.919149e-02	68	86	1.429662e+03	1.284139e-13	7.232874e-05	1.057078e-03

Iter	KKT error	$N_{<10^{-6}}$	$N_{<10^{-2}}$	$\sigma_{\max}$	$\sigma_{\min}$	$\max \delta z_{1:144} $	$\max \delta z_{145:} $
63	7.919148e-02	68	86	1.429675e+03	1.284149e-13	6.030803e-05	8.905660e-04
64	7.919148e-02	68	86	1.429686e+03	1.284157e-13	5.041604e-05	7.504001e-04
65	7.919148e-02	68	86	1.429696e+03	1.284164e-13	4.223115e-05	6.323739e-04
66	7.919148e-02	68	86	1.429704e+03	1.284170e-13	3.542958e-05	5.329647e-04
67	7.919148e-02	68	86	1.429710e+03	1.284175e-13	2.975855e-05	4.492189e-04
68	7.919148e-02	68	86	1.429716e+03	1.284179e-13	2.501783e-05	3.786569e-04
69	7.919147e-02	68	86	1.429721e+03	1.284183e-13	2.104685e-05	3.191955e-04
70	7.919147e-02	68	86	1.429725e+03	1.284185e-13	1.771550e-05	2.690829e-04
71	7.919147e-02	68	86	1.429728e+03	1.284188e-13	1.491745e-05	2.268458e-04
72	7.919147e-02	68	86	1.429731e+03	1.284190e-13	1.256519e-05	1.912439e-04
73	7.919147e-02	68	86	1.429733e+03	1.284192e-13	1.058633e-05	1.612333e-04
74	7.919147e-02	68	86	1.429735e+03	1.284193e-13	8.920725e-06	1.359346e-04
75	7.919147e-02	68	86	1.429737e+03	1.284194e-13	7.518208e-06	1.146073e-04
76	7.919147e-02	68	86	1.429738e+03	1.284195e-13	6.336863e-06	9.662737e-05
77	7.919147e-02	68	86	1.429739e+03	1.284196e-13	5.341575e-06	8.146905e-05
78	7.919147e-02	68	86	1.429740e+03	1.284197e-13	4.502890e-06	6.868929e-05
79	7.919147e-02	68	86	1.429741e+03	1.284197e-13	3.796069e-06	5.791466e-05
80	7.919147e-02	68	86	1.429742e+03	1.284198e-13	3.200317e-06	4.883045e-05
81	7.919147e-02	68	86	1.429742e+03	1.284198e-13	2.698137e-06	4.117135e-05
82	7.919147e-02	68	86	1.429743e+03	1.284199e-13	2.274808e-06	3.471373e-05
83	7.919147e-02	68	86	1.429743e+03	1.284199e-13	1.917931e-06	2.926908e-05
84	7.919147e-02	68	86	1.429744e+03	1.284199e-13	1.617063e-06	2.467846e-05
85	7.919147e-02	68	86	1.429744e+03	1.284200e-13	1.363406e-06	2.080789e-05
86	7.919147e-02	68	86	1.429744e+03	1.284200e-13	1.149548e-06	1.754442e-05
87	7.919147e-02	68	86	1.429744e+03	1.284200e-13	9.692416e-07	1.479281e-05
88	7.919147e-02	68	86	1.429745e+03	1.284200e-13	8.172199e-07	1.247277e-05
89	7.919147e-02	68	86	1.429745e+03	1.284200e-13	6.890450e-07	1.051661e-05
90	7.919147e-02	68	86	1.429745e+03	1.284200e-13	5.809752e-07	8.867253e-06
91	7.919147e-02	68	86	1.429745e+03	1.284200e-13	4.898562e-07	7.476575e-06
92	7.919147e-02	68	86	1.429745e+03	1.284200e-13	4.130290e-07	6.304007e-06
93	7.919147e-02	68	86	1.429745e+03	1.284200e-13	3.482516e-07	5.315337e-06
94	7.919147e-02	68	86	1.429745e+03	1.284201e-13	2.936340e-07	4.481726e-06
95	7.919147e-02	68	86	1.429745e+03	1.284201e-13	2.475825e-07	3.778852e-06
96	7.919147e-02	68	86	1.429745e+03	1.284201e-13	2.087536e-07	3.186211e-06
97	7.919147e-02	68	86	1.429745e+03	1.284201e-13	1.760144e-07	2.686516e-06
98	7.919147e-02	68	86	1.429745e+03	1.284201e-13	1.484098e-07	2.265189e-06
99	7.919147e-02	68	86	1.429745e+03	1.284201e-13	1.251346e-07	1.909939e-06
100	7.919147e-02	68	86	1.429746e+03	1.284201e-13	1.055097e-07	1.610403e-06